Evolution of Cosmic Structure

Lecture 16 - Galaxy classes and scaling relations

Suhail Dhawan

UNIVERSITY OF BIRMINGHAM

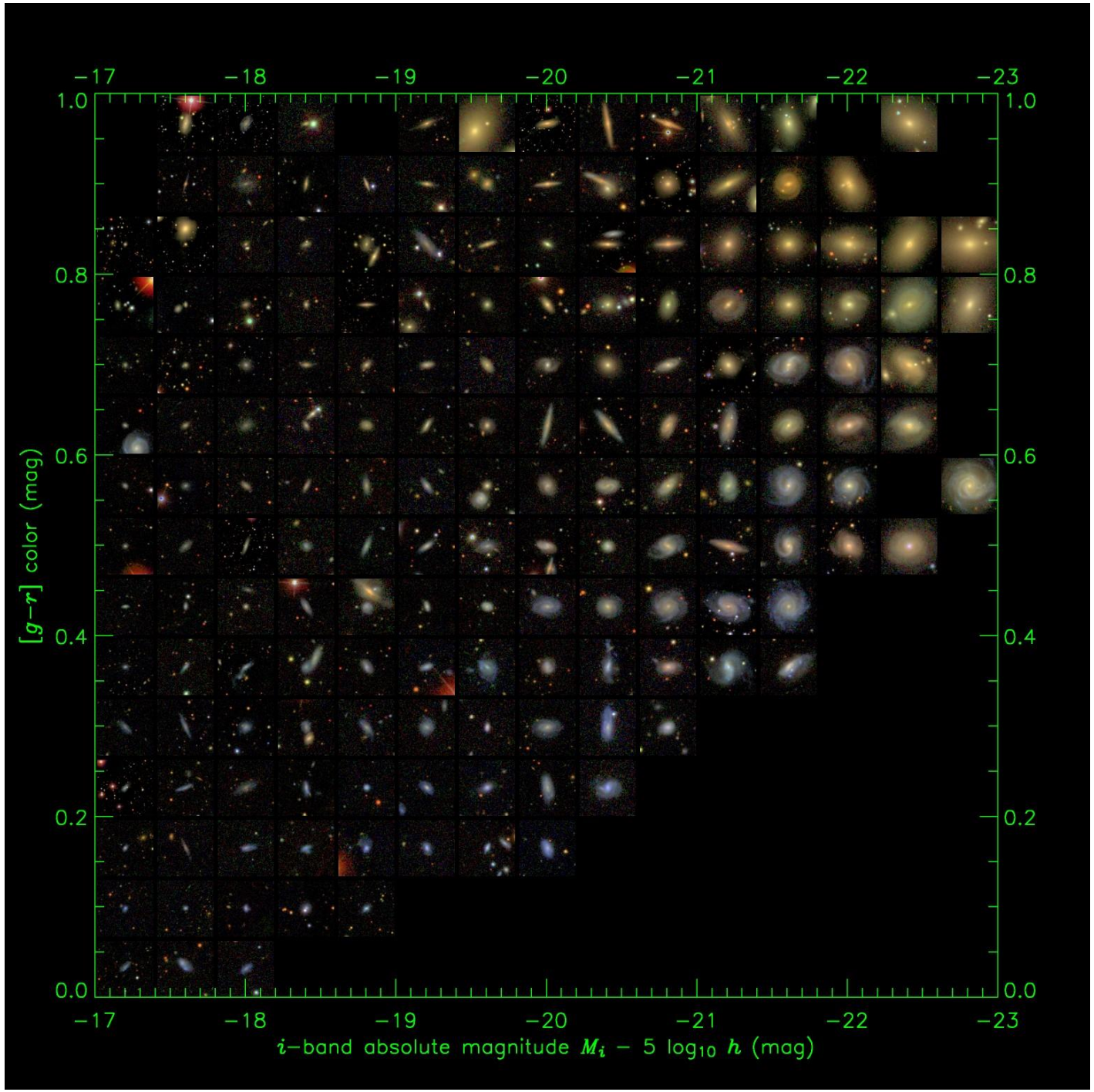
s.dhawan@bham.ac.uk

How can the zoo of galaxies be usefully classified, and what key physical characteristics differentiate the main classes?

What is the impact of hierarchical structure development on galaxies?

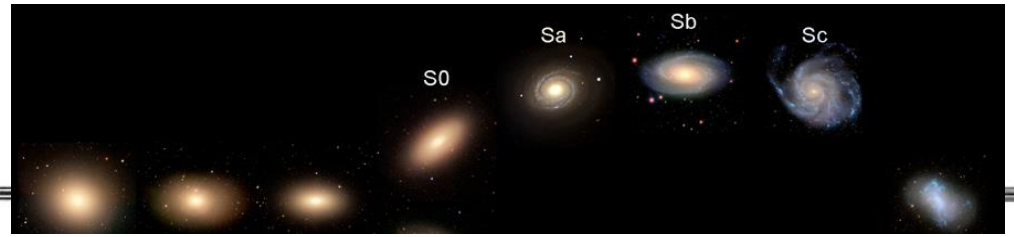
What are the main scaling properties of galaxies, and what do they tell us?

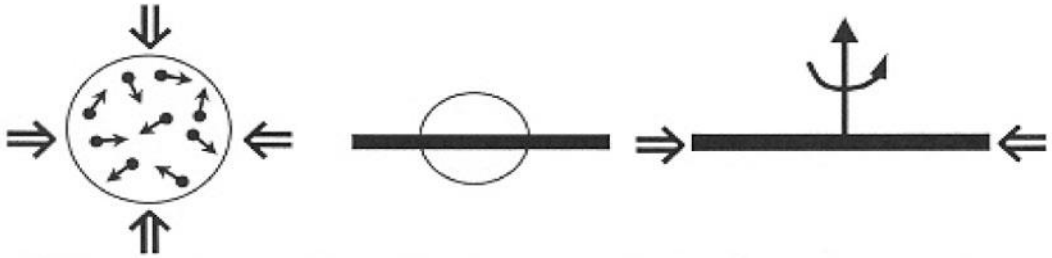
- The physical characteristics of galaxies
- Galaxy scaling relations
  - ✧ Tully-Fisher
  - ✧ Faber-Jackson and the fundamental plane
  - ✧ Black hole mass



The zoo of galaxies, arranged in colour and luminosity. Note that the brightest galaxies tend to be rather yellow.

Galaxy types





Ellipticals	Lenticulars	Spirals	Irregulars
spheroid		**disk**	
"thermal" support		centrifugal support	
slow rotation		rapid rotation	
redder color		**bluer color**	
old pop's		old+young pop's	
higher Z's		lower Z's	
coeval formation		extended formation	
less gas		more gas	
low SFR		high SF	
compact		extended	
more massive		less massive	
20%	10%	70%	

redder

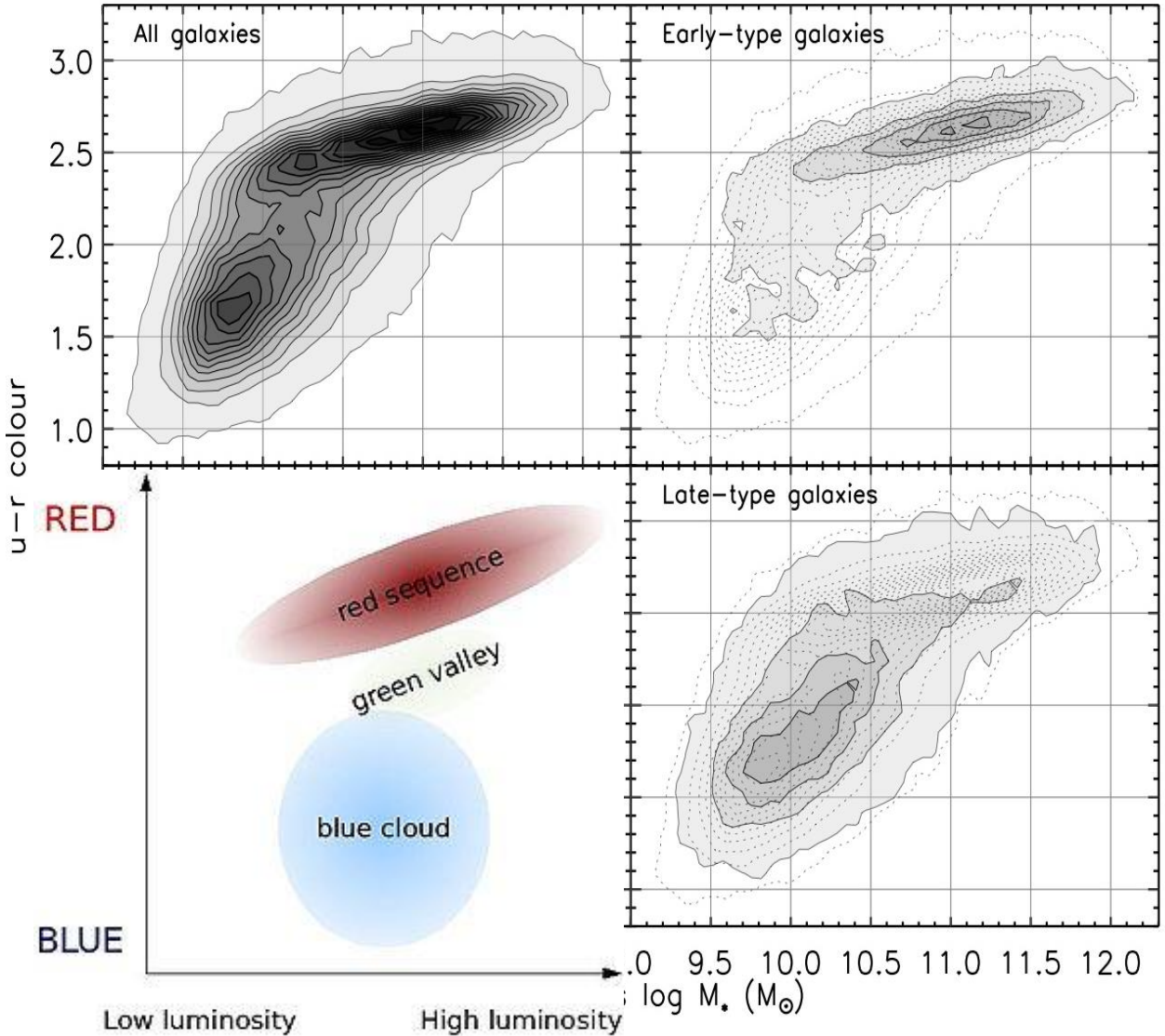
bluer

Although the distinction between barred and unbarred spirals does not seem to be fundamental. Many galaxy properties do progress in a rather well-defined way along the sequence from late to early-type galaxies.

We would like to understand these patterns in galaxy properties in terms of the way they have formed.

Avila-Reese (2006)

Galaxy types



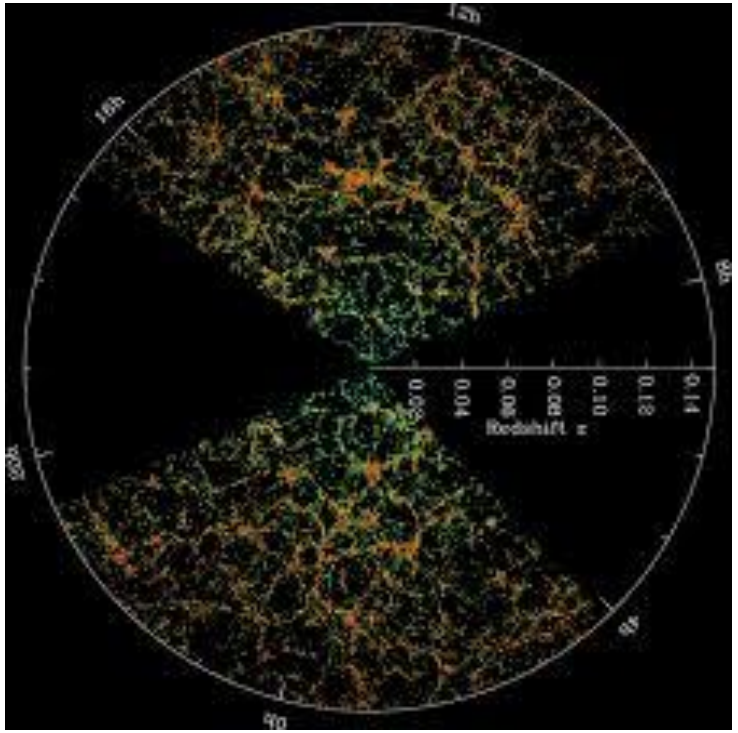
The two main concentrations of galaxies seen in the colour-luminosity plane are known as the red sequence and blue cloud. Between these is a region now called the green valley, which is more sparsely populated with galaxies.

Galaxy types and their environment

A correlation called the color-density relation has been found, and is important to interpret the large scale structure and learn about galaxy formation

Let us look at the density of the large scale structures

Voids -> filaments -> groups -> clusters



Galaxy types and their environment

Red galaxies are typically seen in denser environments, e.g. 70-90% of the brightest cluster galaxies are the
S4

- Hot intracluster gas removes cold gas which would form stars (Ram Pressure Stripping)

- Frequent high-speed encounters can disturb galaxies, leading to gas loss and disk disruption (Galaxy Harassment)

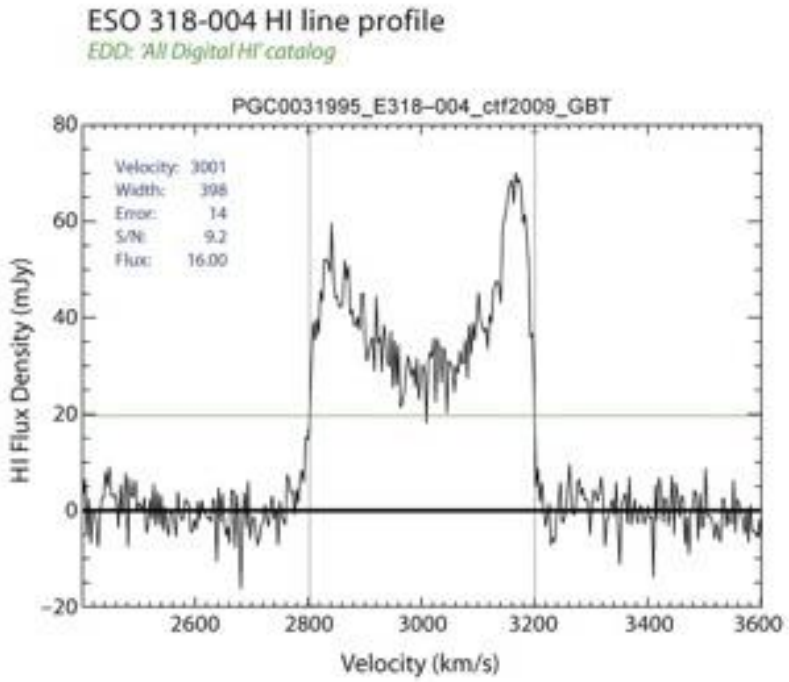
- Supply of new gas can be cut off, meaning all cold gas is used up (Starvation or Strangulation)

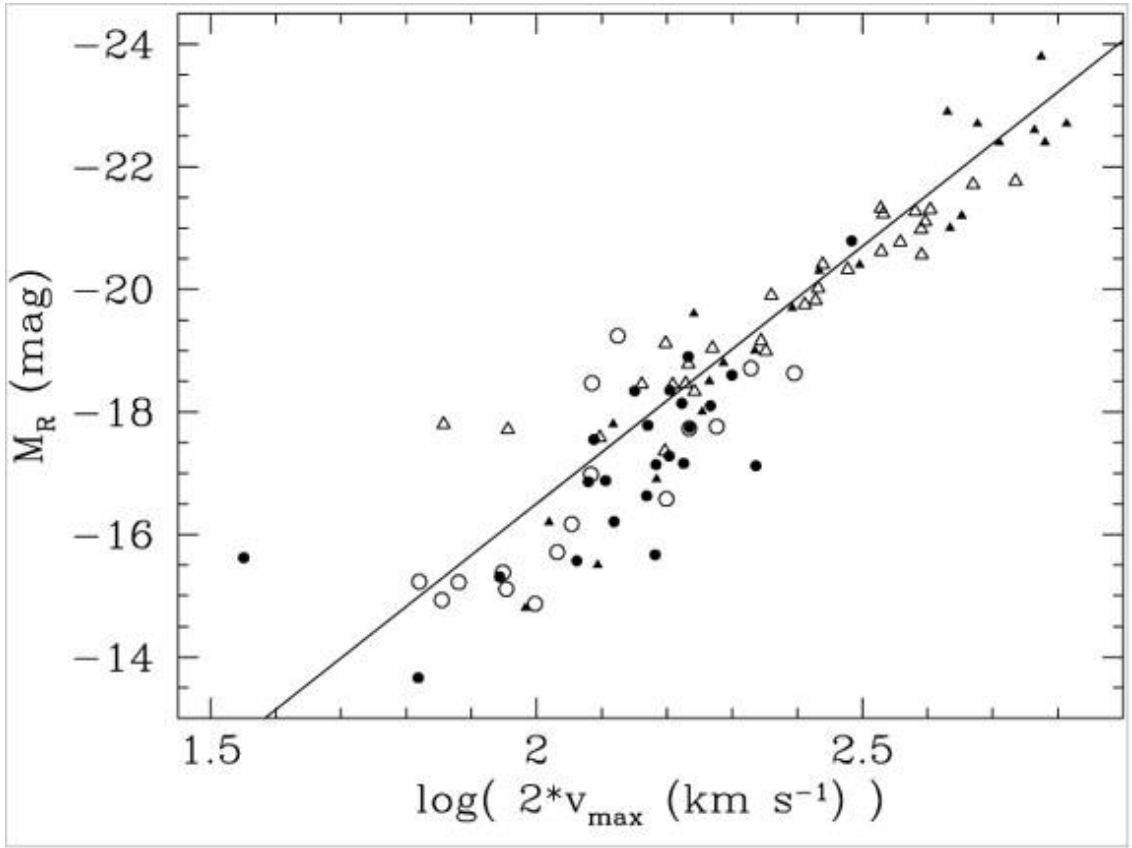
Redder galaxies -> denser environments implies their clustering bias is different from those of bluer galaxies

Recall: $\delta_m = b\ \delta_g$

Galaxy scaling relations

Observed correlations of spiral galaxies



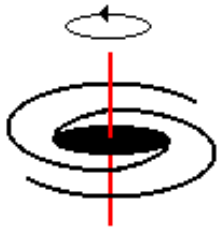


Galaxy scaling relations

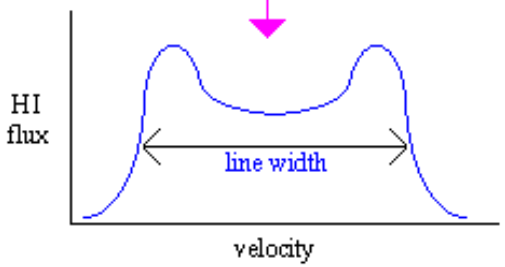
The most important scaling relation for disk galaxies is that between rotation velocity and optical luminosity, known as the Tully-Fisher relation.

The rotation speed can be estimated from Doppler broadening of optical or radio lines – correcting for galaxy inclination.

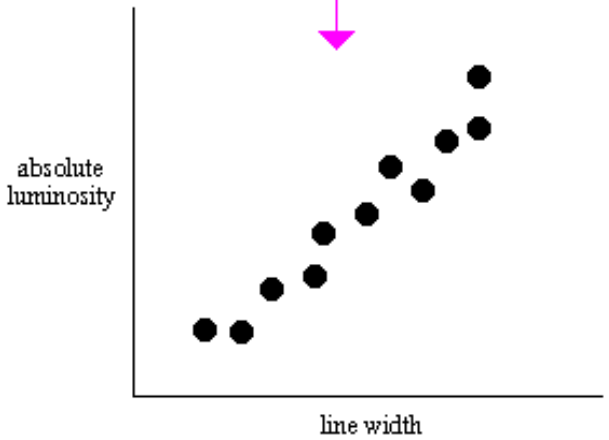
The practical importance of the relation derives from the fact that it can be used to estimate galaxy *distances*.



spiral galaxies rotate, and the rotation speed is proportional to the mass of the galaxy



measurements of neutral hydrogen (HI) display a ''double–horned'' profile, where the width of the line indicates the mass



a plot of line width versus absolute luminosity of a galaxy is called the Tully–Fisher relation. When calibrated using galaxies with Cepheid distances, the TF relation is used to determine Hubble's constant.

Tully-Fisher Relation

Disks.

We know v_c = constant at some point.

This implies mass density falls as r^{-2}.

Just by setting centripetal force equal to gravitational:

Thus $$v_c^2 = \frac{GM(<R)}{R}$$

at R_{200}: $$v_c^2 = \frac{GM_{200}}{R_{200}}$$

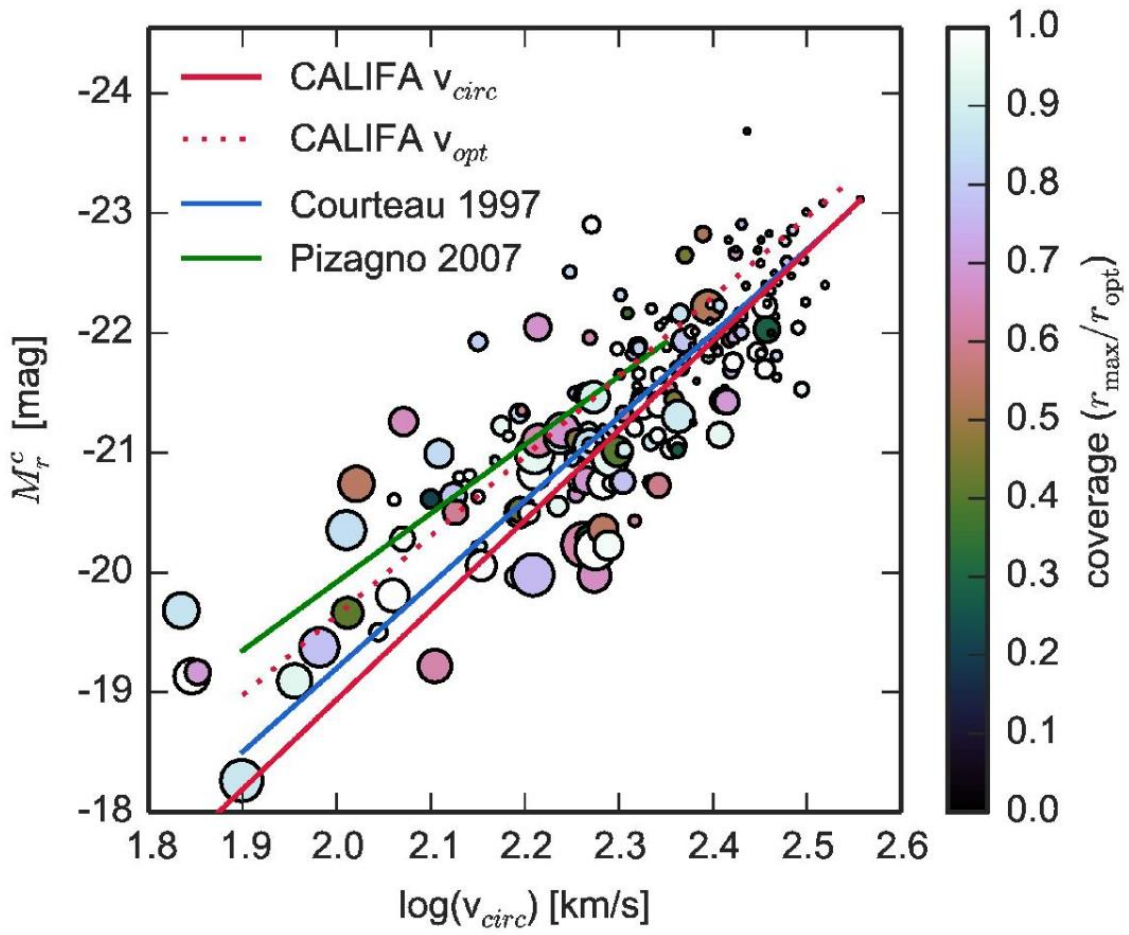
from top-hat model

$$M_{200} = A R_{200}^3$$

so $$v_c \propto M_{200}^{1/3}$$

Or, M goes as v^3

Galaxy scaling relations



Observationally, this is slope found to be ~4. For example, in the I band, one finds (Sparke & Gallagher)

$$L_I = (4\times10^{10}\ L_\odot)(v_{max}/200\ \mathrm{km\ s^{-1}})^4$$

McGaugh et al show that a relationship of this form holds across a wide mass range of galaxies.

Tully-Fisher Relation

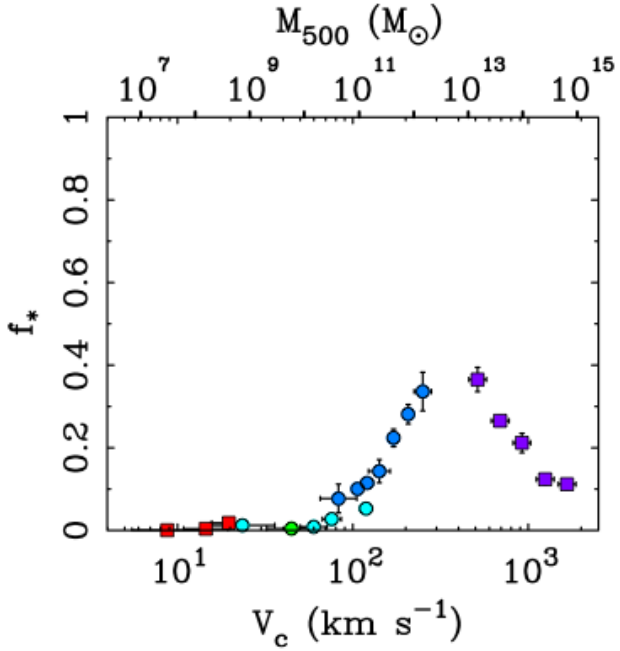
This potentially means something else varies with M_{200},

e.g. $M_{200} \propto v_c^3$

$$M_{200} \frac{L}{M_{200}} \propto v_c^3 \frac{L}{M_{200}}$$

$$L \propto v_c^3 \frac{L}{M_{200}}$$

$\frac{L}{M_{200}} \propto v$ more luminosity per mass at high v



Galaxy scaling relations

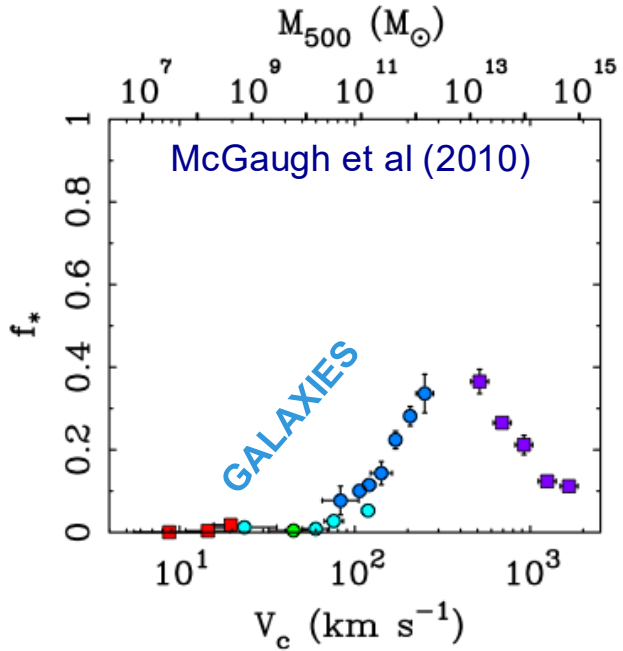
Since the rotation velocity curve is quite flat in spirals, we can relate v_c to the total mass of the system in a rather simple way. It is easy to show that v_c=constant implies a mass density profile which falls off with radius as r^{-2}. The mass within radius r, then simply increases linearly with radius. It then follows that $v_c^2=GM(<R)/R$. If we evaluate this at r_{200}, for example, we get $v_c^2=GM_{200}/r_{200}$.

From the top-hat collapse model, we expect that all systems which virialise at a given epoch should have the same mean density.

So $M_{200}=Ar_{200}^3$ and therefore v_c scales as $M_{200}^{1/3}$.

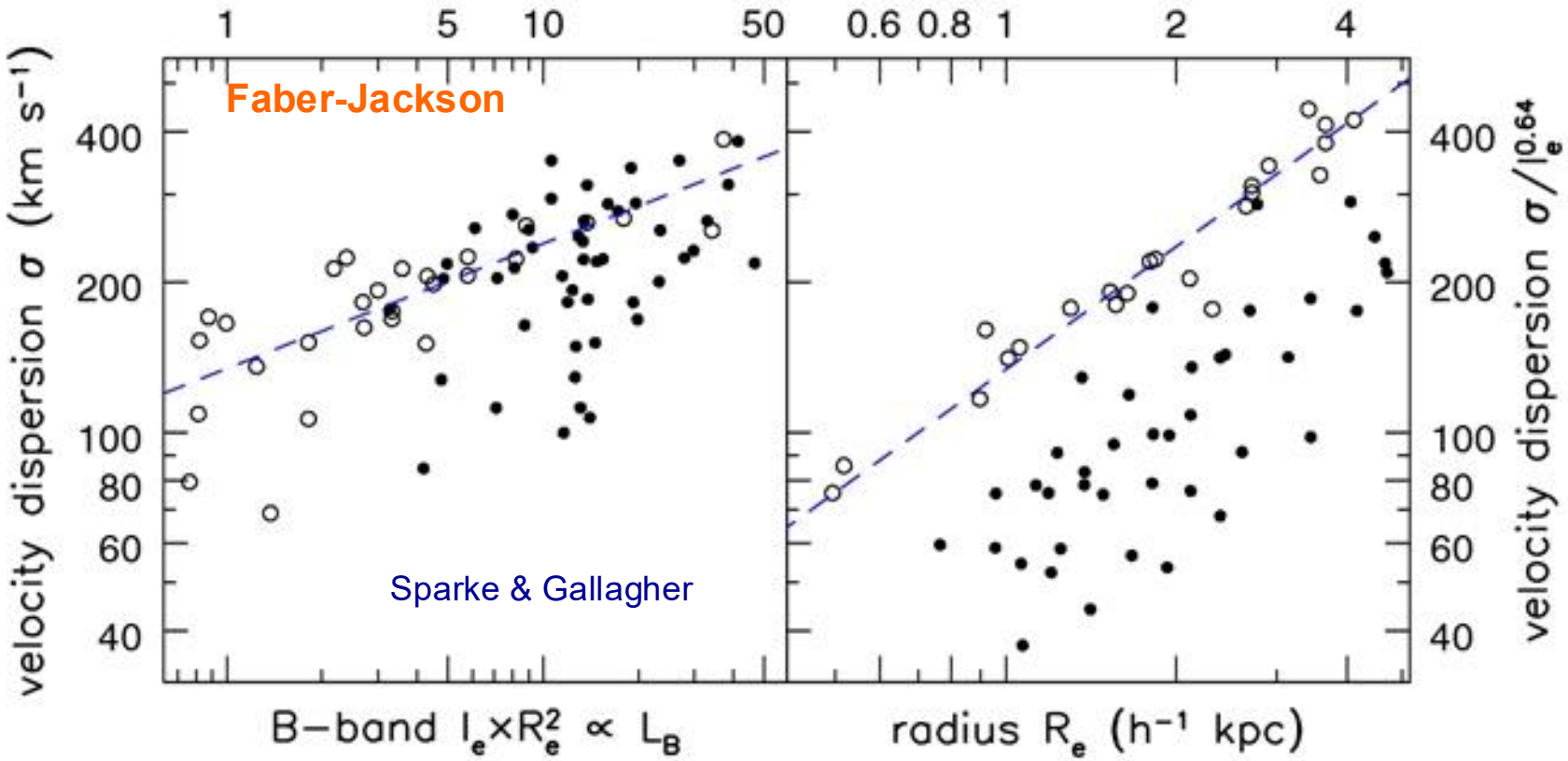
Hence M_{200} should scale as v_c^3, and since we observe that stellar luminosity scales as v_c^4, this implies that the luminosity-to-mass ratios of disk galaxies must rise roughly proportionately to v_c (or equivalently, as $M_{200}^{1/3}$).

This can be seen in McGaugh's plot of stellar mass fraction (i.e. M_*/M_{tot}), which rises through the galaxy regime, before falling again in clusters. It seems that the efficiency of star formation peaks for halos with $M\sim10^{12}M_\odot$.



Galaxy scaling relations

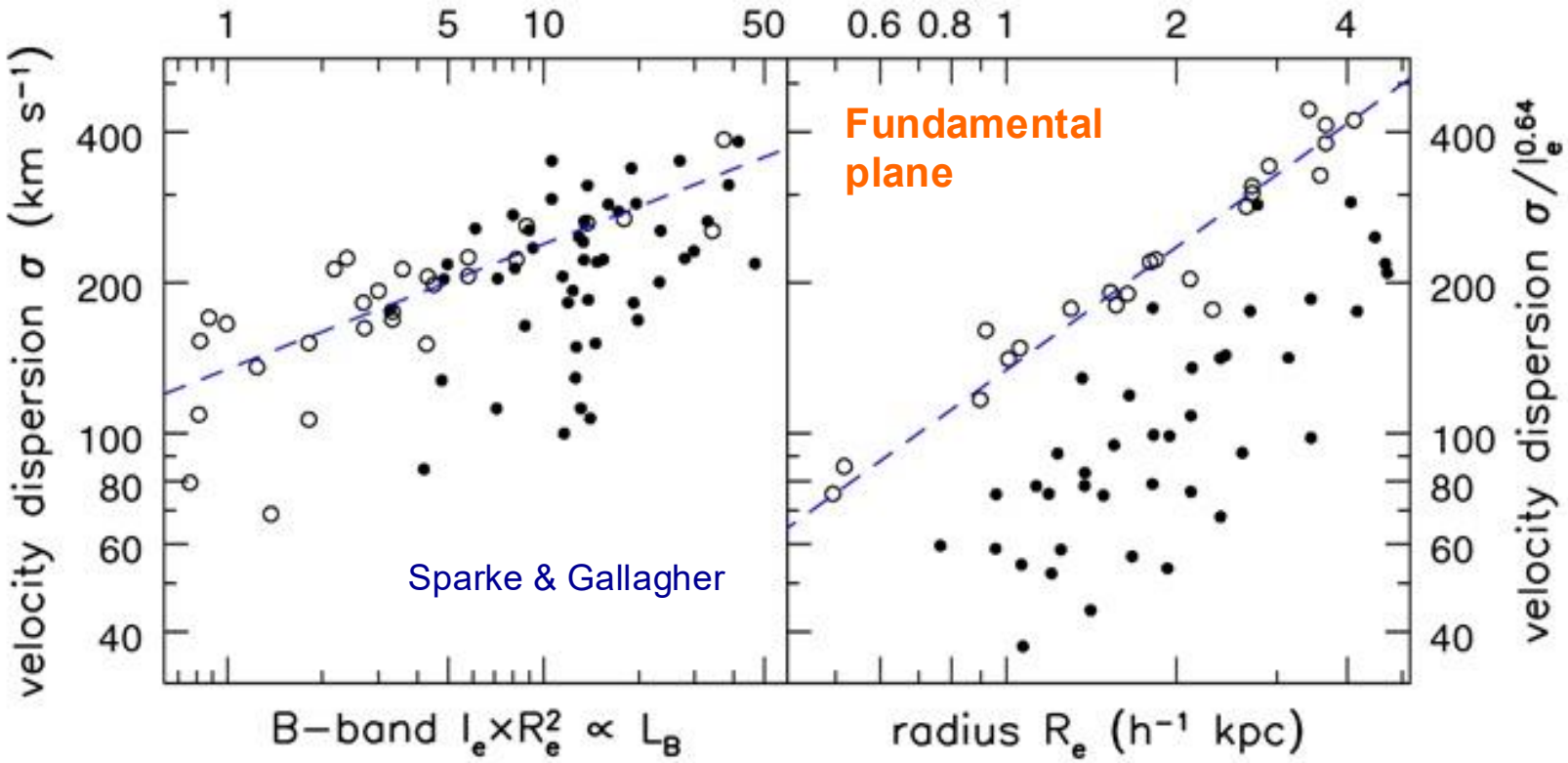
Early-type galaxies also exhibit scaling properties. The analogous relation to the spiral T-F scaling is the Faber-Jackson relation, which says that total optical luminosity scales with the internal velocity dispersion (which may be estimated from line widths) as $L \alpha \sigma^4$.



Circles denote galaxies in the Coma cluster. Dots are galaxies at z~1.

Galaxy scaling relations

However, early-type galaxies also follow a tighter relation involving internal velocity dispersion, effective radius r_e (i.e. the radius containing half the light) and the mean surface brightness, I_e, within r_e , of the form: $r_e \propto \sigma^{1.2} I_e^{-0.8}$, such that, $\log r_e = A + 1.2 \log \sigma - 0.8 \log I_e$ where the factors 1.2 and 0.8 vary somewhat from study to study. This is known as the Fundamental Plane (FP) of ellipticals.



Circles denote galaxies in the Coma cluster. Dots are galaxies at z~1.

Fundamental Plane

Where does this come from?

Virial theorem

$$n \frac{G M M_*}{R} = 2 \frac{M_* 3\sigma^2}{2}$$

or (dividing by M_*)

$$\sigma^2 R \propto M \propto L \frac{M}{L} \propto I R^2 \frac{M}{L}$$

or $\quad r_e \propto \sigma^2 I_e^{-1} \frac{L}{M}$

$$\log r_e = A + 2 \log \sigma - \log I_e$$

$$\boxed{\frac{L}{M} \propto I_e^{0.2} \sigma^{-0.8}}$$

tilt of the fundamental plane.

Galaxy scaling relations

How can we understand this surprisingly tight 3-parameter relationship?

The virial theorem requires that a stellar system in equilibrium should have its gravitational binding energy balance by twice its kinetic energy – i.e.

$$\eta \frac{GMM_*}{R} = 2\frac{M_* 3\sigma^2}{2} \ ,$$

where η depends upon the spatial arrangement of dark matter and stars, σ is the 1D (line-of-sight) velocity dispersion of stars, and M is the <u>total</u> gravitational mass.

Dividing through by M_* we see that

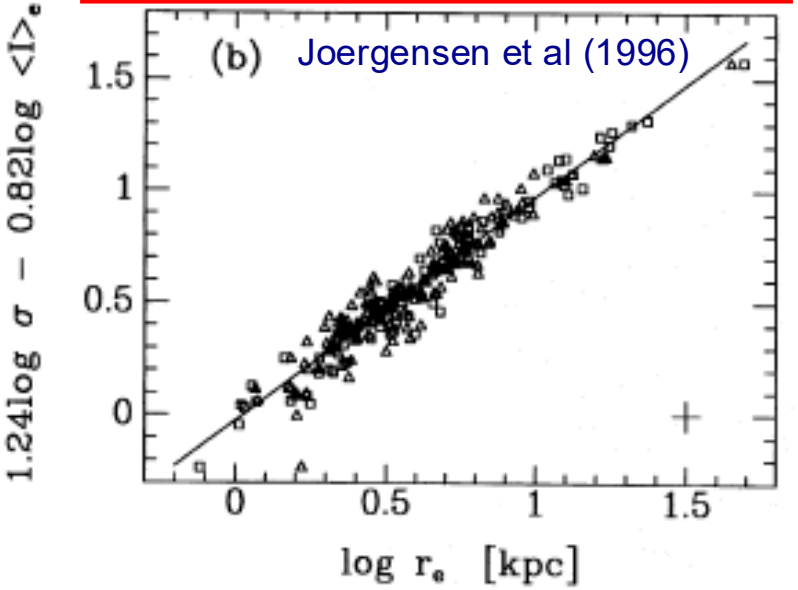
$$\sigma^2 R \propto M \propto L\frac{M}{L} \propto IR^2\frac{M}{L} \, .$$

Hence if both η and (M/L) were independent of system mass, the relationship between the three FP parameters would be

$$\log r_e = A + 2\log\sigma - \log I_e \, .$$

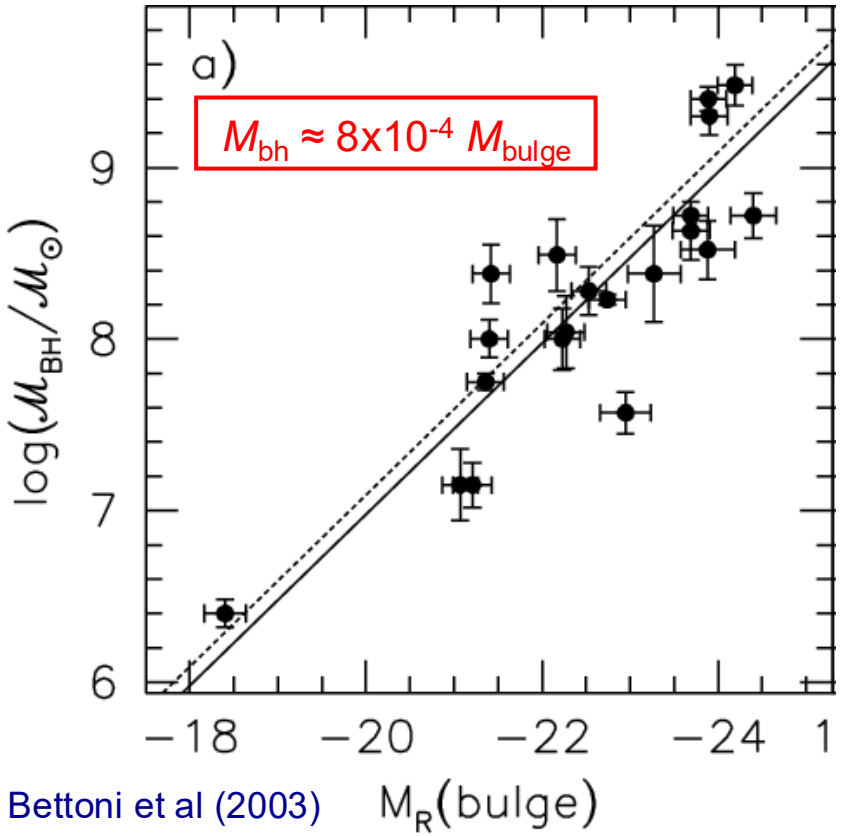
The difference between this and the observed relation (the "tilt" of the FP) can be understood as arising from systematic variation in radial structure &/or M/L with galaxy mass.

$$\log r_e = A + 1.2\log\sigma - 0.8\log I_e$$

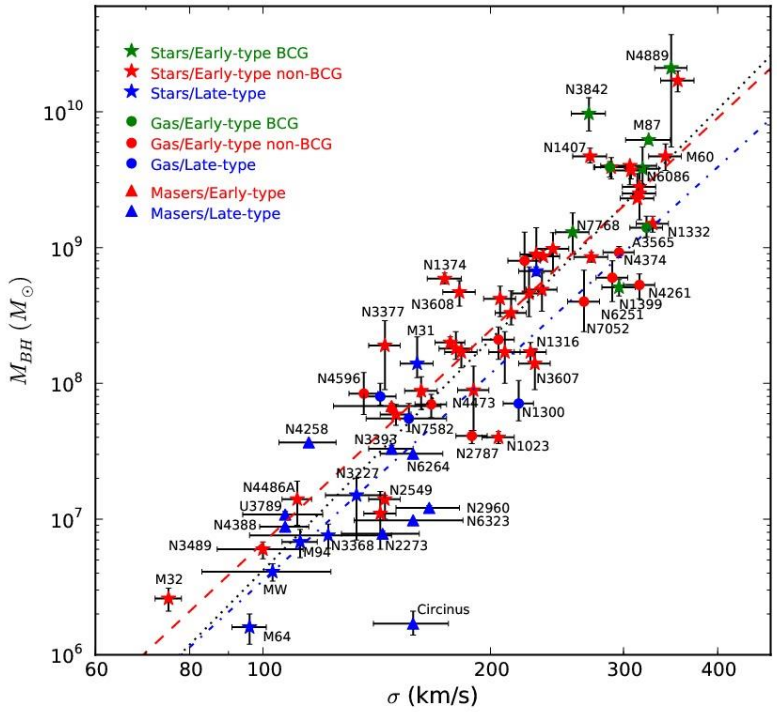


Galaxy scaling relations

We conclude by noting that surprisingly tight relations have been discovered between the masses of central supermassive black holes, determined from studies of the central dynamics of nearby galaxies, and the luminosity and velocity dispersion of the bulges of galaxies of all Hubble types. These relations demonstrate that the growth of stellar bulges and black holes must be related.



Bettoni et al (2003)



BH Scaling relation

BH scaling

- assume isothermal sphere

Eddington luminosity - balancing radiation and gravity

Radiation pressure

$$P_{rad} = \frac{L}{c} \frac{1}{4\pi R^2}$$

force is

$$F_{rad} = P_{rad} \kappa M$$

BH Scaling relation

This needs to balance gravity

$$F_{rad} = F_g$$

$$\frac{L}{c}\frac{1}{4\pi R^2}\kappa M = \frac{G M m_{BH}}{R^2}$$

So Eddington Luminosity is.

$$L_{edd} = \frac{4\pi G m_{BH} c}{\kappa}$$

BH Scaling relation

$$\rho = \frac{\sigma^2}{2\pi G r^2}$$

Now, weight of the gas

$$M = 4\pi \int_0^R \rho r^2 dr = 4\pi \int_0^R \frac{\sigma^2}{2\pi G} dr = \frac{2\sigma^2 R}{G}$$

if gas Mass is $M_g = f_g M$ so

$$F = \frac{G M_* M_g}{R^2} = \frac{4\sigma^4 f_g}{G}$$

if Eddington thrust equals this

$$\frac{L_{Edd}}{c} = \frac{4\sigma^4 f_g}{G}$$

$$M_{BH} = \frac{f_g k}{\pi G^2} \sigma^4$$